

- 2 (a) either $pV = NkT$ or $pV = nRT$ and $n = N / N_A$ C1
 clear correct substitution e.g.
 $2.5 \times 10^5 \times 4.5 \times 10^3 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 290$ M1
 $N = 2.8 \times 10^{23}$ A0 [2]
 (allow 1 mark for calculation of $n = 0.467 \text{ mol}$)
- (b) (i) volume = $(1.2 \times 10^{-10})^3 \times 2.8 \times 10^{23}$ or $\frac{4}{3} \pi r^3 \times 2.8 \times 10^{23}$ C1
 $= 4.8 \times 10^{-7} \text{ m}^3$ $2.53 \times 10^{-7} \text{ m}^3$ A1 [2]
- (ii) either $4.5 \times 10^3 \text{ cm}^3 \gg 0.48 \text{ cm}^3$ or ratio of volumes is about 10^{-4} B1
 justified because volume of molecules is negligible B1 [2]

[Total: 6]

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2 (a) (i) two objects of different masses at same temperature (M1)